

Commutative Algebra Solutions

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This is a complete solution set to all the problems in Andreas Gathmann's commutative algebra script, as well as alternative proofs of some of the theorems. The script itself is sort of a more modern version of Atiyah Macdonald. Solutions are brief and generally aim to give the main ideas.

As a sidenote, I did these problems after reading Atiyah-Macdonald in a 4 week frenzy for a class. Even despite these circumstances, it might be my favorite classic textbook of all time. Nonetheless, I find the exposition in Gathmann's script to be on average better due to more geometric intuition and typesetting (thanks L^AT_EX). Moreover, while there are less exercises, they are very good exercises which I enjoyed and feel like the right questions to ask.

Introduction

0.15

- (a) A nonzero polynomial cannot have infinitely many roots. Therefore $\mathcal{I}(X) = 0$ and $X = \mathbb{A}$, a contradiction.
- (b) Suppose that $f \in \mathcal{I}(X)$. Then when we take $f(-, 0)$ we get a polynomial with infinite roots. Therefore it must be the 0 polynomial. However then it is 0 on the entire X , axis, a contradiction.
- (c) Suppose that $f \in \mathcal{I}(X)$. Then when we take $f(-, 0)$ we get a polynomial with infinite roots. Therefore it must be the 0 polynomial. However then it is 0 on the entire X , axis, a contradiction.
- (d) Note that any polynomial in 1 variable has only finitely many roots. Then use the same argument.
- (e) It is certainly true that some of the things of this form are varieties. Therefore we need to construct a counterexample. Take the polynomial map $\mathbb{R} \rightarrow \mathbb{R}$ which is defined by x^2 . The image is all the real numbers ≥ 0 . Therefore use the same argument as (a).

0.16

- (a) This is because the product of the two highest order terms has order at most the sum of the orders.
- (b) If R is an integral domain then the product of the two highest order terms will be nonzero and the equality holds. Otherwise take $ab = 0$ and $(ax)(bx) = 0$ as a counterexample.

Ideals

1.8

One direction is clear. I will prove this by induction on n , so write $J_k = I_1 \cap \cdots \cap I_k$. I claim that J_{n-1} is coprime to I_n . If this is the case write

$$a + b = 1 \quad a \in J_{n-1} \quad b \in I_n$$

Then take $ax + bx = x$ so that $x \in J_{n-1}I_n$ and $J_n = J_{n-1}I_n$. Now imply induction on J_{n-1} .

Now I need to prove the claim. By the inductive hypothesis J_{n-2} , I_{n-1} , and I_n are pairwise coprime. Therefore write

$$\begin{aligned} a + b &= 1 & c + d &= 1 \\ a \in J_{n-2} & & c \in I_{n-1} & & b, d \in I_n \end{aligned}$$

and

$$1 = (a + b)(c + d) = ac \pmod{I_n}$$

which is an element of $J_{n-2} \cap I_{n-1} = J_{n-1}$.

1.9

Consider the quotient module I/I^2 , which is a vector space with dimension n . If I had fewer generators, then this vector space would have dimension less than n , a contradiction.

1.11

- (a) The subvarieties of $\mathbb{A}_k^1$ over an algebraically closed field are exactly any finite set of points. We can check that each of these varieties $k_1, \dots, k_n$ is defined by the radical ideal $((x - k_1) \cdots (x - k_n))$. Furthermore all radical ideals of $\mathbb{C}[x]$ have this form, which can be checked using the PID property.
- (b) the ideal defined by $x^2 + 1$ is radical since this is an irreducible polynomial. On the other hand no element of $\mathbb{R}$ satisfies it so the variety it defines is just the 0 variety.

1.13

Note that the generators of the LHS are the pairwise products of the generators of the RHS. Therefore by 1.8 it suffices to show that the two ideals on the RHS are comaximal. Take

$$(y - 1)^2 + 2y - y^2 = 1$$

as desired.

1.19

- (a) Suppose that $a \in I$. Then $\phi(a) \in I^e$ so $a \in (I^e)^c$.
- (b) Suppose that $a \in (I^e)^e$. Therefore $a = \sum r_i \phi(a_i)$ where $r_i \in R$ and $a_i \in I^c$. Therefore $\phi(a_i) \in I$ and $a \in I$.
- (c) Both of these ideals are generated by all elements of the form $\phi(i)\phi(j)$ for $i \in I$ and $j \in J$.
- (d) Mapping into an intersection is equivalent to mapping into both ideals.

1.20

First we need to establish a basic fact. This is that taking the image of a map of varieties corresponds to taking a contraction of the coordinate rings. This is because the polynomials which vanish on the image are exactly those whose pullback vanishes on the domain.

- (a) This says that $(I \cap J)^c = I^c \cap J^c$. Note that the extension is the smallest ideal whose contraction is the thing inside the brackets. However taking the contraction commutes with intersections.

- (b) This says that $I^c + J^c \subseteq (I + J)^c$. Suppose that $\phi(i) \in I$ and $\phi(j) \in J$. Then it is clear that $\phi(i + j) = \phi(i) + \phi(j) \in I + J$.
- (c) This states that $[I^c : J^c] \subseteq [I : J]^c$. Suppose that $\phi(a)J \subseteq I$. Then consider aj for $j \in J^c$, then $\phi(aj) = \phi(a)\phi(j) \in I$. Therefore $aj \in I^c$ and $aJ^c \subseteq I^c$.

1.22

We can prove this theorem using the first isomorphism theorem. Consider the quotient map $R/I \rightarrow R/J$ which is well defined by the universal property of the quotient. The image is all of R/J and the kernel is J/I .

Prime and Maximal Ideals

2.8

We have

$$I_1 = (5, x^3 + 2x + 3) \quad I_2 = (2, x^2 + x + 1)$$

we want to understand whether the quotients or integral domains/fields. The quotients are

$$\mathbb{F}_5[x]/(x^3 + 2x + 3) \quad \mathbb{F}_2[x]/(x^2 + x + 1)$$

we iterate over the roots to see if these polynomials can be factored. It turns out that the second is irreducible (hence a maximal ideal). However the polynomial $x^3 + 2x + 3$ is divisible by $x - 4$ in $\mathbb{F}_5$. Therefore the first ideal is neither prime nor maximal.

2.9

- (a) Suppose that $x^n \in \phi^{-1}(I)$. Then $\phi(x^n) = \phi(x)^n \in I$ so $\phi(x) \in I$ and $x \in \phi^{-1}(I)$.
- (b) Suppose that $ab \in \phi^{-1}(\mathfrak{p})$, then $\phi(a)\phi(b) \in \mathfrak{p}$ so either a or b is in $\phi^{-1}(\mathfrak{p})$.
- (c) Consider the inclusion map $k \rightarrow k[x]$. The ideal (x) is maximal but the preimage is (0) , which is not maximal.

2.10

These are the prime avoidance theorems, which say stuff about the decomposition of a variety into its irreducible component.

- (a) We use the contrapositive. Suppose that $I_k \not\subseteq \mathfrak{p}$. Then take elements $a_k \in I_k \setminus \mathfrak{p}$. Multiplying them all together gives an element $a_1 \cdots a_n$ in the intersection of the I_k without being in the prime ideal P .
- (b) Firstly, we ignore redundant terms in the union and assume WLOG that none of the $\mathfrak{p}_i$ are contained in the union of all the other prime ideals. Therefore take elements $x_i \in \mathfrak{p}_i$ but $x_i \notin \mathfrak{p}_j$ for $i \neq j$. Now use the contrapositive, suppose that $I \not\subseteq \mathfrak{p}_k$. Therefore take elements $y_i \in I \setminus \mathfrak{p}_i$. Now consider the element

$$\sum_i y_i \prod_{j \neq i} x_j = y_i \prod_{j \neq i} x_j \pmod{\mathfrak{p}_i} \neq 0$$

as desired.

- (c) We use contrapositive again. Firstly, take some $a \in I \setminus P_1$ and $b \in P_1 \setminus (\mathfrak{p}_2 \cup \dots \cup \mathfrak{p}_n)$ (using the nonredundance). Let $K \subseteq \{2, \dots, n\}$ be the set of indices so that $a \in \mathfrak{p}_k$ and consider the element

$$a + b \sum_{i \in K} y_i \prod_{j \neq i} x_j$$

Where the $x_2, \dots, x_n$ and $y_2, \dots, y_n$ satisfy the properties in (b). Therefore we have

$$\begin{aligned} a + b \sum_{i \in K} y_i \prod_{j \neq i} x_j &= a \neq 0 \pmod{P_1} \\ a + b \sum_{i \in K} y_i \prod_{j \neq i} x_j &= b y_r \prod_{j \neq r} x_j \neq 0 \pmod{\mathfrak{p}_r} \quad r \in K \\ a + b \sum_{i \in K} y_i \prod_{j \neq i} x_j &= a \neq 0 \pmod{\mathfrak{p}_r} \quad r \notin K \end{aligned}$$

2.11

- (a) We claim the the quotient by this ideal is k . Indeed we have the map $R \rightarrow k$ defined by evaluation at a . This is surjective (any constant function is continuous), and the kernel is I_a by definition.
- (b) This function will not be 0 on any point in the interval. Therefore its inverse is well defined on the interval.
- (c) Suppose for the contrapositive that $V(I) = \emptyset$. Then take f_a for every $a \in [0, 1]$ such that $f_a(a) \neq 0$. Let U_a be an open set centered containing a such that f_a is nonzero on U_a . Since the interval is compact we may cover it with finitely many of these open sets $a_1, \dots, a_n$. Therefore take $f_{a_1}^2 + \dots + f_{a_n}^2$ which has full support and is therefore invertible. We are done.
- (d) This is by (a), which establishes that there are continuous functions of this form, and by (c), which implies that all maximal ideals are of this form.

2.12

- (a) It suffices to show that any domain with the property is a field (since quotienting by a prime ideal gives a domain). Note that $a^n = a$ implies that $a^{n-1} = 1$ in a domain so $a^{n-2} = a^{-1}$. Therefore every element is invertible and we are done.
- (b) Take $\mathbb{Z}/(2) \times \mathbb{Z}/(2)$.

2.23

- (a) We can consider the quotient R/I . Therefore it suffices to show that every ring has a minimal prime ideal. This is by Zorn's lemma, order the prime ideals by reverse inclusion. The upper bound of a chain of smaller and smaller prime ideals is their infinite intersection.
- (b) Any such prime ideal must contain x^2y . Therefore it must contain x or y . However (x) and (y) are prime ideals, therefore these are the minimal ones.

2.24

Take $a \in P$ and $b \in Q$ so that $a + b = 1$. Then raise this equation to the power of $n + m$ and we see that every term in the expansion is in P^n or Q^m by pigeonhole. However the RHS is just 1 and we are done.

2.25

- $a \Rightarrow b$ Suppose that x is not a unit, then it is contained in a maximal ideal, which is prime and therefore is $\mathfrak{p}$. Then x is contained in the intersection of all prime ideals and is nilpotent.
- $b \Rightarrow c$ The radical of (0) is all the nilpotent elements. If all not-nilpotent elements are units then this ideal is maximal.
- $c \Rightarrow a$ If the intersection of all prime ideals is maximal then there can only be one ideal in the intersection. Otherwise take one of the terms.

2.26

- (a) Consider the poset of all sets of countably infinite disjoint subsets of M . Order the sets by containment, then apply zorns lemma. The upper bound is a union, which is well defined.
- (b) Write M is a disjoint union of countably infinite subsets. $M = \bigcup A_i$. We need a bijection

$$A_i \rightarrow A_i \times A$$

at which point the bijection $M \rightarrow A \times M$ follows. This is by the diagonal argument.

2.27

- (a) Consider the poset of sets of linearly independent elements, ordered by containment. The upper bound of a chain is well defined as the infinite union. Therefore apply Zorn's lemma.
- (b) We can $\mathbb{R}^n$ and $\mathbb{R}^m$ as vector spaces over $\mathbb{Q}$. We can use (a) to construct the hamel basis over $\mathbb{Q}$. Therefore by 2.26a $\mathbb{R}$ is isomorphic to $(\bigcup R_i) \times \mathbb{Q}$ where R_i is a countably infinite index subset of the hamel basis. Note that $\mathbb{Q}$ is countably infinite so we may apply 2.26b to give the isomorphism, which is a group isomorphism by the construction.

Modules

3.5

- (a) We can lift a generating set for M/N into M via representatives. We can then take generators for N inside M . Together these will generate M .
- (b) A generating set for M will generate M/N when projected into the quotient.
- (c) Let $R = k[x_1, x_2, \dots]$. Then R is a finitely generated module over itself (generated by 1). However $I = (x_1, x_2, \dots)$ is not since I/I^2 is an infinite dimensional vector space over R/I .

3.11

- $a \Rightarrow b$ This is by the lattice theorem.
- $b \Rightarrow c$ Choose any element $m \in M/N$. Note that Rm is a submodule so it must be the whole thing. Then $M/N \cong R/\text{Ann}(m)$. By the condition that there are no intermediate ideals $\text{Ann}(m)$ must be maximal.
- $c \Rightarrow a$ There are no ideals between a maximal ideal and the ring. In this context a submodule would correspond to such an ideal.

3.16

Suppose that I is principal, generated by x . Then the map $R \rightarrow I$ defined by $r \mapsto rx$ is an isomorphism since it is an integral domain.

Now suppose that I is a free R module with basis $a_1, \dots, a_n$. If there are more than 1 elements in the basis then note that $a_2 \cdot a_1 - a_1 \cdot a_2 = 0$, a contradiction. Therefore there is one element in the basis, which generates I as an ideal.

3.19

(a) Suppose that

$$0 = M_0 \subset \dots \subset M_n = M$$

is a composition series for M . If $N \subset M$ then we may take the series

$$0 = M_0 \cap N \subset \dots \subset M_n \cap N = N$$

Now consider the quotients

$$\frac{M_{k+1} \cap N}{M_k \cap N} \subseteq \frac{M_{k+1}}{M_k}$$

therefore all these terms are either simple or trivial. Then by ignoring redundant terms we obtain a smaller composition series for N .

(b) Suppose that we have alternative composition series

$$0 = M_0 \subset \dots \subset M_m = M$$

$$0 = M'_0 \subset \dots \subset M'_n = M$$

with $n \geq m$. We will prove the theorem by induction on m . Suppose that $M_{m-1} = M'_{n-1}$, then the result $m = r$ follows from the inductive hypothesis. Otherwise take the series.

$$0 = M'_n \cap M_0 \subset \dots \subset M'_{n-1} \cap M_{m-1} \subset M'_{n-1} \subset M$$

And then consider k to be the minimal index so that $M_k \not\subseteq M'_{n-1}$. Then we have

$$\frac{M_k \cap M'_{n-1}}{M_{k-1} \cap M'_{n-1}} \subset \frac{M_k}{M_{k-1}}$$

which must be 0 since it is a proper subset. Therefore we may remove this term. Now we have a composition series of length m starting with M'_{n-1} . Therefore by the inductive hypothesis $n = m$.

(c) It suffices to construct composition series between M_k and M_{k-1} . This is equivalent to constructing a composition series for M_k/M_{k-1} . We may construct a series for M_k by (a). Then quotient all the terms by M_{k-1} and remove everything that becomes trivial. This will be a valid composition series by diamond theorem.

3.21

The lengths of $\mathbb{Z}_8$ and $\mathbb{Z}_{12}$ are 3. The length of $R/(x)$ when R is a PID is the number of terms in the UFD irreducible decomposition of x .

3.29

- (a) The maximal ideal is all the elements whose numerator is divisible by p . This is because all the other elements are invertible.
- (b) We can achieve any a/b by taking $(p/1) \cdot a/(bp)$, since $p/1$ is an element of I . However 1 is not an element of this ideal since it is an ideal.
- (c) There is no finite set of generators, we must have infinitely large powers of p in the denominators. A countably infinite set of generators is $1, 1/p, 1/p^2, \dots$

Exact Sequences

4.9

- (a) Consider the meaning of the exact sequence

$$M_1 \xrightarrow{\phi_1} M_2 \xrightarrow{\phi_2} M_3 \rightarrow 0$$

which states the ϕ_1 sends M_1 surjectively into the kernel of ϕ_2 . Therefore the maps which are killed by ϕ_1^* are exactly those which kill this kernel. These are the exactly the maps which factor through ϕ_2 (this can be shown), IE those in the image of ϕ_2^* .

Now suppose that the sequence

$$0 \rightarrow \text{Hom}(M_3, N) \xrightarrow{\phi_2^*} \text{Hom}(M_2, N) \xrightarrow{\phi_1^*} \text{Hom}(M_1, N)$$

is exact for all N . Consider the category of pairs $(\psi : M_2 \rightarrow N, N)$ such that $\psi \circ \phi_1 = 0$. This condition says that (ϕ_2, M_3) is an initial object in this category, therefore unique. Then the condition exactly determines the pair ϕ_2, M_3 up to isomorphism. At this point we are done since $M_1 \xrightarrow{\phi_1} M_2 \rightarrow M_2/\ker(\phi_1) \rightarrow 0$ is exact and satisfies the above.

- (b) Consider the injective map:

$$\mathbb{Z} \rightarrow \mathbb{Z} \quad n \mapsto 2n$$

Now we have

$$\text{Hom}(\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \rightarrow \text{Hom}(\mathbb{Z}, \mathbb{Z}/2\mathbb{Z})$$

via precomposition by multiplication by 2. The image is 0 and this is not surjective.

4.16

- (a) The maps are

$$a \mapsto (a, -a) \quad (a, b) \mapsto a + b$$

which is easy to check is exact.

- (b) Consider the map of exact sequences

$$\begin{array}{ccccccc} 0 & \longrightarrow & I \cap J & \longrightarrow & I \oplus J & \longrightarrow & I + J \longrightarrow 0 \\ & & \downarrow a & & \downarrow b & & \downarrow c \\ 0 & \longrightarrow & R & \longrightarrow & R \oplus R & \longrightarrow & R \longrightarrow 0 \end{array}$$

and apply snake lemma to see the exactness of the sequence

$$\ker(c) \rightarrow \text{coker}(a) \rightarrow \text{coker}(b) \rightarrow \text{coker}(c) \rightarrow 0$$

which after some manipulation becomes:

$$0 \rightarrow R/(I \cap J) \rightarrow R/I \oplus R/J \rightarrow R/(I + J)$$

- (c) Suppose that $R = k[x, y]$ and $I = (x)$ and $J = (y)$. Therefore suppose that there is a retraction ψ of $\phi : (xy) = I \cap J \rightarrow I \oplus J = (x) \oplus (y)$. Then consider $a = (x, 0)$ and $b = (0, x)$. We can write

$$y(a - b) = (xy, -xy) \implies y\psi(a - b) = \psi(y(a - b)) = xy$$

however $\psi(a)$ and $\psi(b)$ are in (xy) and therefore the result must be in (xy^2) . This is a contradiction.

Now suppose that there is a section ψ of $\phi : R/(x) \oplus R/(y) \rightarrow R/(x, y)$. This will be a map $k \rightarrow k[x] \oplus k[y]$. Suppose that $1 \mapsto (a, b)$. Then we need $1x \mapsto (ax, 0)$ however x acts by 0 on k so we need $a = 0$, by the same argument we need $b = 0$, a contradiction.

4.17

It suffices to show that

$$0 \rightarrow N \rightarrow M \xrightarrow{\phi} R^n \rightarrow 0$$

is split exact. Indeed, all free R modules are projective. This is we can because by surjectivity we can choose elements $m_i \in M$ which maps to $(\underbrace{0, \dots, 0}_{i-1 \text{ times}}, 1, 0, \dots, 0)$. This determines the section by the universal property of the direct product.

Tensor Products

5.9

We claim that both of these satisfy the analogous universal property of the tensor product for trilinear maps. At this point uniqueness follows from the Yoneda lemma. Therefore consider the diagram and suppose that γ is a trilinear map.

$$\begin{array}{ccccc} L \times M \times N & \xrightarrow{\beta_1} & L \times (M \otimes N) & \xrightarrow{\beta_2} & L \otimes (M \otimes N) \\ & & \searrow \gamma & \searrow \phi_1 & \downarrow \phi_2 \\ & & & & P \end{array}$$

A map out of $L \times (M \otimes N)$ is a map out of L and $M \otimes N$, where the latter factors uniquely through $M \otimes N$ by bilinearity. Then apply bilinearity again to see that the map $L \times (M \otimes N) \rightarrow P$ factors uniquely through $L \otimes (M \otimes N)$ as desired.

We can make the analogous argument to show that $(L \otimes M) \otimes N$ satisfies this universal property.

5.11

For $\mathbb{Q}/\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z} = 0$. This is because $\frac{a}{b} \otimes \frac{c}{d} = \frac{a}{1} \otimes \frac{c}{db} = 0 \otimes \frac{c}{db} = 0$

Note that $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q} = \mathbb{Q}$. This is because any $\mathbb{Z}$ -linear map out of $\mathbb{Q} \times \mathbb{Q}$ factors through $\mathbb{Q}$ where the map is multiplication. This can be proven by noting that the image of any $\frac{a}{b} \otimes \frac{c}{d}$ can be shown to be the image of $\frac{ac}{bd} \otimes 1$.

Note that $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} = \mathbb{C} \times \mathbb{C}$. We can write

$$\begin{aligned}\mathbb{C} \otimes \mathbb{C} &\cong \mathbb{R}[x]/(x^2 - 1) \otimes \mathbb{C} \\ &\cong \mathbb{C}[x]/(x^2 - 1) \\ &\cong \mathbb{C}[x]/(x - i) \times \mathbb{C}[x]/(x + i) \\ &\cong \mathbb{C} \times \mathbb{C}\end{aligned}$$

Note that

$$\mathbb{Q}[x] \otimes_{\mathbb{Q}} \mathbb{C} = \mathbb{C}[x]$$

5.12

- (a) Consider the large rectangle. We can chop up the rectangle along the lines formed by all the edges of the triangles. This forms a large grid of much smaller rectangles. Then summing them up two different way we either get $a \otimes b$ or $\sum a_i \otimes b_i$.
- (b) Suppose that every rectangle has a side with rational length. Then when we sum up we can organize the terms into something of the form $x \otimes 1 + 1 \otimes y$. If we pass the the quotient by $\mathbb{Q}$ this is 0. Therefore $a \otimes b$ must be zero in the quotient $\mathbb{R}/\mathbb{Q} \otimes \mathbb{R}/\mathbb{Q}$. Therefore at least one of a, b must be rational.

5.13

- (a) All the maps are given in the problem, the formal verification that these are well defined maps is left to ChatGPT. They are basis independent because these maps are defined without respect to any basis.
- (b) Suppose that $x = \sum f_i \otimes w_i$ where we have chosen $w_1, \dots, w_n$ to be linearly independent. Now suppose that $\phi(x) = 0$. Therefore for any $v \in V$ we have $\sum f_i(v)w_i = 0$. In particular, each $f_i(v)w_i$ must be 0 by linear independence. Therefore each f_i is 0 and we are done.
- (c) The spaces have the same dimension, therefore since they are injective they are isomorphisms.
- (d) This is the trace map.

5.18

The equation that we need to show is that

$$(M \otimes_R N) \otimes_R R' = (M \otimes_R R') \otimes_{R'} (N \otimes_R R')$$

note that the proof of Exercise 5.9 shows that the tensor product is associative even when tensoring over various rings (as long as they have a compatible action). Therefore rearrange the brackets in the RHS to yield:

$$(M \otimes_R R') \otimes_{R'} (N \otimes_R R') = (M \otimes_R N) \otimes_R (R' \otimes_{R'} R') = (M \otimes_R N) \otimes_R R'$$

Coproduct Tensor Product

We claim that in the category of R -algebras, the tensor product is the coproduct. This means that an R -algebra map $A \otimes B \rightarrow C$ corresponds uniquely to a pair of R -algebra maps $A \rightarrow C$ and $B \rightarrow C$. This is by the correspondance

$$\phi(-, -) \mapsto \phi(-, 1) \times \phi(1, -) \quad f \times g \mapsto f(-)g(-)$$

which are inverses by the homeomorphism property:

$$\phi(a \otimes b) = \phi((a \otimes 1) \cdot (1 \otimes b)) = \phi(a \otimes 1) \cdot \phi(1 \otimes b)$$

5.21

This problem is nicer if you read proposition 5.22 first.

- (a) We have shown that the tensor product is right exact. Therefore we have

$$R/I \otimes_R R/J \cong \frac{R \otimes_R R/J}{I \otimes_R R/J} \cong \frac{R/J}{IR/J} \cong R/(I+J)$$

- (b) Consider the universal property of the product $X \times Y$

$$\begin{array}{ccccc} & & Z & & \\ & \swarrow f_X & \downarrow f & \searrow f_Y & \\ X & \xleftarrow{\pi_X} & X \times Y & \xrightarrow{\pi_Y} & Y \end{array}$$

which says that f is uniquely determined by two functions f_X and f_Y . Note that maps of varieties correspond to maps of coordinate rings in the opposite direction. Therefore if we reverse all the arrows we get the universal property of the coproduct.

5.24

- (a) Suppose that the sequence is split exact. Then we have $M_2 \cong M_1 \oplus M_3$. Therefore $M_2 \otimes N = (M_1 \otimes N) \oplus (M_3 \otimes N)$. Now we can reconstruct the exact sequence

$$0 \rightarrow (M_1 \otimes N) \rightarrow M_2 \otimes N \rightarrow (M_3 \otimes N) \rightarrow 0$$

by noting that the induced maps after the tensor product are still the projection maps.

- (b) Suppose that we have $N = R^n$. $M \otimes N = (M \otimes R)^n = M^n$. Therefore tensoring with N gives the exact sequence:

$$0 \rightarrow M_1^n \rightarrow M_2^n \rightarrow M_3^n \rightarrow 0$$

where the maps are componentwise the original maps.

5.27

Construct two composition series for M and N as

$$M_0 \subset \cdots \subset M_n$$

$$N_0 \subset \cdots \subset N_r$$

Now consider the series formed by filling in the gaps between

$$M_0 \otimes N_r \subseteq M_1 \otimes N_r \subseteq \cdots \subseteq M_n \otimes N_r$$

This series will have length

$$\sum l \left(\frac{M_{k+1} \otimes N_r}{M_k \otimes N_r} \right) = \sum l \left(\frac{M_{k+1}}{M_k} \otimes N_r \right)$$

therefore consider the sequence

$$\frac{M_{k+1}}{M_k} \otimes N_0 \subseteq \frac{M_{k+1}}{M_k} \otimes N_1 \subseteq \cdots \subseteq \frac{M_{k+1}}{M_k} \otimes N_r$$

where we claim that all the quotients are either trivial or simple. Write

$$\frac{\frac{M_{k+1}}{M_k} \otimes N_{i+1}}{\frac{M_{k+1}}{M_k} \otimes N_i} \cong \frac{M_{k+1}}{M_k} \otimes \frac{N_{i+1}}{N_i}$$

so that each quotient is the tensor product of simple R modules, therefore simple or trivial by 5.21. Then the length of each small sequence is at most r . Therefore the maximal sum we get in the big sum is at most nr as desired.

Localization

The theory is nicer if you use the direct limit definition of localization. We write the localization of a module as a direct limit of R -modules:

$$S^{-1}M = \varinjlim_{s \in S} M_s \quad M_s \cong M$$

The maps are as follows, if $ab = c$ then the map $M_a \rightarrow M_c$ is by multiplication by b . In this way, the inverse of $s \in S$ sends $r \in M_a$ to $r \in M_{as}$.

Now we define the universal property of localization for an R -module. Suppose S acts invertably on N , then maps $S^{-1}M \rightarrow N$ are in bijective correspondence with maps $M \rightarrow N$. The proof is by the universal property of the direct limit. That is, an R -module map $\phi : \varinjlim M_s \rightarrow N$ is specified by a compatible system of maps $\phi_s : M_s \rightarrow N$. This means that if $ab = c$ then the diagram must commute:

$$\begin{array}{ccc} M_a & \xrightarrow{\phi_a} & N \\ \downarrow b & \nearrow \phi_c & \\ M_c & & \end{array}$$

We set $a = 1$ in this diagram, then $\phi_c(cm) = \phi_1(m)$ so by invertability of c , ϕ_c is determined by ϕ_1 . Therefore defining $\phi_s(m) = \phi_1(m)/s$ gives the unique compatible system generated by ϕ_1 .

6.12

Note that R is an R module, therefore localize it using the direct limit definition and give it a ring structure, (we have $(a \in R_s) \cdot (b \in R_{s'}) = ab \in R_{ss'}$).

Consider some ring map $R \rightarrow R'$. Then R' becomes an R -module via restriction of scalars. Therefore this map can also be thought of as a module map. In particular the criterion that S is mapped to invertable elements in R' means that S acts invertably on R' . Now apply the universal property of localization for R modules to obtain the unique map. This map is well defined as a map of rings since all we need to check is that 1 maps to 1.

6.13

- (a) Note that P contains y (multiply the generator by y), therefore the quotient of R by Q will just be $k[x, y]/(x - 1, y) \cong k$ which is a field. Furthermore localizing R at P will make x invertable, therefore we have $(xy) = (y)$ and the expression follows.

The fact that P is a maximal ideal means that it corresponds to a point of the affine variety $xy = 0$ (that is $(1, 0)$). The localization means that the functions on this variety which are well defined around that point are isomorphic to the functions on the line well defined at 1.

- (b) One can prove this using the universal property of localization and the universal property of the quotient. A map out of that ring is specified by a map $R \rightarrow R'$ and an image of x , which is uniquely determined as the inverse of a .

The geometric interpretation gives that localizing at an element gives you a variety in a larger space. This variety will be contained in $X \times [-\infty, \infty]$, however when projected down to X it will be $X \setminus Z$ where Z are the zeros of a .

6.14

- (a) This is by Zorn's lemma and then the correspondence theorem of prime ideals
- (b) The ideal found in (a) may not be maximal, it is only guaranteed to be maximal among those ideals not intersecting S . A counterexample is the ring $\mathbb{Z}_{(2)}$, the ring of fractions with odd denominators. Choose the ideal (0) , this is contained in the only maximal ideal. On the other hand if you localize at an even element you get the rationals, whose only prime ideal is (0) .

6.15

We claim that localizing at S or $\bar{S}$ gives the same universal property. Therefore we wish to show that S being invertible is equivalent to every element of $\bar{S}$ being invertible. In one direction note that $S \subseteq \bar{S}$. In the other direction suppose that $s \in \bar{S}$ so that $as \in S$. Then as is a unit if S is invertible so s must be a unit and we are done.

6.16

We don't actually need to assume that M is local, rather than I is contained in the Jacobson radical.

- (a) Suppose that $IM = M$. Then by Nakayama lemma we can find an element $x \in I$ that acts by identity on M . Then $1 - x$ will be an invertible element (since x is in the Jacobson radical) killing M and we are done.
- (b) Let N be the module generated by the m_i . The condition essentially says that $M/IM = N/(IM)$. Therefore $N + IM = M$ and $I(M/N) = (IM)/N = M/N$. Now by (a) we have that $M/N = 0$ so that $M = N$.

6.24

- (a) Note that localizing commutes with the direct sum operation, then consider the image of the natural map

$$M_1 \oplus M_2 \rightarrow M$$

- (b) Recall the exact sequence

$$0 \rightarrow M_1 \cap M_2 \rightarrow M_1 \oplus M_2 \rightarrow M_1 + M_2 \rightarrow 0$$

and localize it, the localization distributes on the last two terms by what we have already done so it must on the first.

Consider $R = \mathbb{Z}$, $S = 2^k$, $M_i = 2^i\mathbb{Z}$ and write

$$0 = S^{-1} \bigcap M_i \neq \bigcap S^{-1} M_i = S^{-1} \mathbb{Z}$$

- (c) Suppose that $a/b \in S^{-1}\sqrt{I}$. Then $a^n \in I$ and therefore $a/b \in \sqrt{S^{-1}I}$. Now suppose that $a/b \in \sqrt{S^{-1}I}$. Therefore $a^n/b^n \in S^{-1}I$ so $a^n \in I$ and $a/b \in S^{-1}\sqrt{I}$.

6.29

- (a) Consider the meaning of $J \subseteq I$. This means that we can form the exact sequence

$$0 \rightarrow J \rightarrow I \rightarrow I/J \rightarrow 0$$

so that the result follows from locality of exactness.

- (b) Recall that the nilradical is $\sqrt{(0)}$. Now apply the result that localizing commutes with taking radicals and that being 0 is a local property.
- (c) Consider the ring $\mathbb{Z}/(6)$. Note that this is not an integral domain. Now suppose that we localize at a prime ideal. The prime ideals are $(2), (3)$. The prime ideals in the localization will correspond to the prime ideals of $\mathbb{Z}$ containing (6) but contained in $\mathfrak{p}$. There is only one such ideal in every case and so when we localize we always get a ring with a unique prime ideal. However we can check again by computation that $\mathfrak{p}(\mathbb{Z}/(6))_{\mathfrak{p}} = 0$ so that the ring is an integral ideal.

Chain Conditions

7.3

- (a) Suppose that N is a submodule of N . Then choose some $a/p^n \in N$ with the highest power of p in the denominators. Then we can solve the equation $ab \cong 1 \pmod{p^n}$ via elementary number theory (or Hensel's lemma). Therefore $1/p^n \in N$ and $\langle 1/p^n \rangle \subseteq N$. Furthermore $N \subseteq \langle 1/p^n \rangle$ since every element with a lower power of p is in $\langle 1/p^n \rangle$.
- (b) The proper submodules are totally ordered by inclusion from (a).

$$\langle 1/p \rangle \subset \langle 1/p^2 \rangle \subset \dots$$

We can see that this is an ascending chain. However there is no descending chain since every descending chain must start at a finite point in the chain.

7.12

Suppose FTSOC that ϕ is not an isomorphism. Then take some element $x \notin \text{im } \phi$. Note that

$$\phi^n(x) \in \text{im } \phi^n \quad \phi^n(x) \notin \text{im } \phi^{n+1}$$

since otherwise we could write

$$\phi^n(x) = \phi^{n+1}(y) \implies x = \phi(y)$$

by injectivity, but we said $x \notin \text{im } \phi$. Therefore the images of ϕ^n form a nonstabilizing descending chain, a contradiction.

7.22

- (a) If a is irreducible then we are done. Otherwise write $a = bc$ where neither a or b are units and then repeat the process on b and c . By the noetherian condition that ideals generated by the terms in the decomposition must stabilize and the process will terminate. We essentially have a tree of ideals.
- (b) Note that $\sqrt{I}$ is generated by finitely many elements $a_1, \dots, a_n$. Let n_i be such that $a_i^{n_i} \in I$. Then $(\sqrt{I})^{\sum n_i} \in I$ by the pigeonhole principle.

7.23

Note that every ideal of $S^{-1}R$ is of the form $IS^{-1}R$ where I is an ideal of R . Then the Noetherian condition follows directly.

7.24

- (a) Let $m_1, \dots, m_n$ generate M as an R module. Consider the map $\phi : R \rightarrow M^n$ by $r \mapsto (rm_1, \dots, rm_n)$. Now take an ascending chain of ideals of R

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \dots$$

which can be extended to an ascending chain of submodules in M^n

$$\phi(I_1) \subseteq \phi(I_2) \subseteq \phi(I_3) \subseteq \dots$$

which must eventually stabilize. However one can show that if $\phi(I) = \phi(I')$ then the ideals are the same after quotienting by the annihilator. This is because they act the same on M .

- (b) We use the same argument as (a) to prove the analogous statement for the Artinian condition. Now note that $R/\text{Ann}(M)$ is Artinian, hence Noetherian. Therefore we have a finitely generated module over a Noetherian ring. Therefore it is Noetherian and we are done.

Prime Factorization and Decomposition

8.8

- (a) Note that $\mathbb{R}[x, y]/(y - x^2) \cong \mathbb{R}[x]$ which is a UFD. Furthermore $R[x, y]/(xy - 1) \cong R[x, x^{-1}]$. This is the localization of a UFD at one element, which remains a UFD by 8.1a (the elements either stay prime or become units).
- (b) This one is quite tricky. We may guess that functions on the unit circle correspond to functions of θ . Therefore consider

$$x \mapsto \frac{e^{i\theta} + e^{-i\theta}}{2} \quad y \mapsto \frac{e^{i\theta} - e^{-i\theta}}{2i} \quad e^{i\theta} \mapsto \cos(\theta)x + \sin(\theta)y$$

which induces an isomorphism

$$\mathbb{C}[x, y]/(x^2 + y^2 - 1) \cong \mathbb{C}[e^{i\theta}, e^{-i\theta}]$$

The RHS is a UFD by the second example in (a).

8.22

- (a) Consider what happens when we lift I into $R[x, y]$. We get $(x^2, (y - 1)(y + 1))$. If we draw the variety correspond to this ideal there are 3 lines which intersect at $(0, 1), (0, -1)$. Therefore the prime ideals containing this are $(x, y - 1)$ and $(x, y + 1)$ (which are maximal). Then note that $(x^2, y - 1) \cap (x^2, y + 1)$ is a suitable primary decomposition by the UFD property. In the quotient these corresponds to simply $(y - 1) \cap (y + 1)$.
- (b) Note that $\mathbb{Z}[\sqrt{5}i] \cong \mathbb{Z}[x]/(x^2 + 5)$. Therefore we need a primary decomposition of $(6, x^2 + 5)$ in $\mathbb{Z}[x]$. Note that this ideal is $(2, x^2 + 5) \cap (3, x^2 + 5)$. These decompose as $(2, (x + 1)^2) \cap (3, x - 1) \cap (3, x + 1)$. These correspond to the ideals $(2) \cap (3, \sqrt{-5} + 1) \cap (3, \sqrt{-5} - 1)$.
- (c) This is a UFD and all the generating elements are irreducible therefore the ideal is (xy, xz, y^2, zy) . Note that the radical ideal of this is (y, xz) which is contained in the prime ideals (y, x) , (y, z) , and (x, y, z) . Therefore consider the primary decomposition

$$(x, y) \cap (y, z) \cap (x, y, z)^2$$

8.29

First we claim that the ideal quotient of a primary ideal by an element is primary. By 8.24 we consider the case where $a \notin \mathfrak{q}$. Then suppose that $ab \in [\mathfrak{q} : x]$. Therefore $abx \in \mathfrak{q}$. This means that either $ax \in \mathfrak{q}$ or $b \in \sqrt{\mathfrak{q}}$. IE either $a \in [\mathfrak{q} : x]$ or $b \in \sqrt{\mathfrak{q}} = \sqrt{\mathfrak{q}} : x$ (by 8.24).

Now suppose that I is an ideal in a Noetherian ring. By the proof of 8.27 choose some a so that $[I : a]$ is $\mathfrak{p}$ -primary. If $[I : a] \neq \mathfrak{p}$ then we may find some x with $x \notin [I : a]$ but $x^n \in [I : a]$ (with n minimal). Therefore take $[I : ax^{n-1}]$ since $x^{n-1} \notin [I : a]$ this ideal is still $\mathfrak{p}$ -primary, however it contains x and therefore strictly contains $[I : a]$. If we repeat this process we get an ascending chain of ideals. The only way this stabilizes is if we can eventually no longer pick such an element. IE $[I : a] = \mathfrak{p}$.

8.32

- (a) Recall the condition for a UFD that every element f is a product of elements which are prime $f = p_1 \cdots p_n$. Suppose that $\mathfrak{p}$ is a minimal prime over f . Then one of the p_i must be contained in $\mathfrak{p}$. Therefore since $(p_i) \subseteq \mathfrak{p}$ we have $(p_i) = \mathfrak{p}$ by minimality.

In the other direction take a minimal prime ideal lying over (r_1) . It is principle so we have $(r_1) \subseteq (p)$. Therefore $r_1 = p_1$ for some a . Now repeat the argument to get a chain of ideals $(r_1) \subseteq (r_2) \subseteq (r_3) \subseteq \cdots$. This process must terminate by the Noetherian condition so we obtain the factorization $r_1 = p_1 \cdots p_n$.

- (b) Suppose that R is a UFD. A minimal non-zero prime ideal is a minimal prime ideal lying over the ideal (0) . Now apply (a).

Integral Ring Extensions

9.8

- (a) No it is not integral. Note that $x^4 - 4x + 2$ is the minimal polynomial for $\sqrt{2 + \sqrt{2}}$ and $(2x)^3 - 3$ is the minimal polynomial for $\frac{1}{2}\sqrt[3]{3}$. One of these is monic while the other is not.
- (b) Note that $\mathfrak{p} = \mathfrak{p}' \cap R = (x^2 - 1)$. We want to consider the element $\alpha = \frac{1}{x+1} \in R'$. Suppose that it is integral over $R_{\mathfrak{p}}$. Then we may write:

$$\alpha^n + \alpha^{n-1}\beta_{n-1} + \cdots \beta_0 = 0 \quad \beta_i \in \mathbb{R}[x^2 - 1]_{\mathfrak{p}}$$

so multiply by $(x+1)^n$ to obtain.

$$1 + (x+1)\beta_{n-1} + \cdots (x+1)^n\beta_0 = 0$$

Note that each β_i can be written like

$$\beta_i = \frac{\gamma_i}{1 + f(x^2 - 1)}$$

so that they are well defined at $x = 1$. Therefore consider the expression in $\mathbb{R}[x]_{(x+1)}$, which is a DVR with uniformizer $(x+1)$. The order of the expression in the DVR is 1 by properties of DVRs. Therefore we are done.

This is not a contradiction with 9.5b since in this context S is a multiplicatively closed subset of R' , not of R .

9.12

Note that $\mathbb{R}[t]$ is a UFD, hence normal. Therefore since we know $t \in \overline{R}$ it suffices to show that:

$$R \subseteq R[t] \cong \mathbb{R}[t]$$

The inclusion is obvious, therefore write

$$R[t] \cong \frac{\mathbb{R}[x, y, t]}{(y^2 - x^2 - x^3)}$$

and note that $x = 1 - t^2$ and $y = t(1 - t^2)$. Therefore $R[t]$ is a quotient of $\mathbb{R}[t]$. However in $R[t]$ note that

$$y^2 - x^2 - x^3 = t^2 x^2 - x^2 - x^3 = x^2(t^2 - 1 - (1 - t^2)) = 0$$

Therefore $R[t] \cong \mathbb{R}[t]$ as desired.

9.13

- (a) The problem is asking to prove that integral closure commutes with localization. From 9.7b we know that $S^{-1}\overline{R} \subseteq \overline{S^{-1}R}$ (since that problem says $S^{-1}\overline{R}$ is integral over $S^{-1}R$). Therefore suppose that $x \in \overline{S^{-1}R}$ and write:

$$x^n + \frac{a_{n-1}}{s_{n-1}}x^{n-1} + \cdots + \frac{a_0}{s_0} \quad a_i \in R \quad s_i \in S$$

We can multiply through by the product of the denominators to obtain an expression

$$s'_1 x^n + s'_2 a_{n-1} x^{n-1} + \cdots + s'_0 a_0$$

therefore consider $y = xs'_1$ and write (by plugging in $x = y/s'_1$ and multiplying through by s_1^{n-1}):

$$y^n + s'_1 s'_2 a_{n-1} y^{n-1} + \cdots + s_1^{n-1} s'_0 a_0$$

therefore y is in $\overline{R}$, and therefore x is in $S^{-1}\overline{R}$.

- (b) This problem is just an application of 9.7b. In particular set $R' = \text{Frac}(R)$ and note that localizing a field doesn't do anything.
- (c) Note that the condition of being integrally closed means that there is an isomorphism $R \rightarrow \overline{R}$. Since localization commutes with integral closure the result follows from the fact that a map is an isomorphism if it is locally an isomorphism (at maximal ideals).

9.14

Suppose that $fg \in \overline{R}[t]$. If we can find an element $\alpha \in R'$ which satisfies $fg(\alpha) = 0$, then it follows that $\alpha \in \overline{R}$. Indeed, if we can factor fg , then f, g factor with roots in $\overline{R}$ (since they are roots of fg) so they must both be in $\overline{R}[t]$.

Therefore we would like to pretend that $h = fg$ factors. Suppose that h is irreducible, then we consider the ring $\frac{R'[x]}{h(x)}$ in which h has a root. Otherwise perform this procedure on one of the irreducible factors of h . After doing this some times we obtain a larger ring $R' \subseteq R''$ in which h (and therefore f and g) factor completely.

Finally we have the claim:

Claim 10.0.1. If $A \subseteq B \subseteq C$ then let A_B be the integral closure of A in B and A_C be the integral closure of A in C . Then $A_B = A_C \cap B$.

If $fg \in \overline{R}[t]$, then pass to $R''[t]$ and find the coefficients of f, g to be in the integral closure, then note that they are also in R' so we are done by the claim.

Finally we need to prove the claim. In one direction every element of A_C is integral over A , therefore $A_C \cap B \subseteq A_B$. In the other direction every element of A_B is integral over A in C , so $A_B \subseteq A_C$.

9.22

- (a) Consider the ideal (z, y) . If we quotient by this we get $\mathbb{C}[x]$, therefore this ideal is prime and does not intersect $\mathbb{C}[x]$ except at the identity. However then this ideal is an ideal comparable to 0 lying over 0. This is a contradiction with integrality.
- (b) Yes this extension is integral. This is because y satisfies a monic polynomial in $\mathbb{C}[x]$ and z satisfies a monic polynomial in $\mathbb{C}[x, y]/(y^3 - x^2)$. Therefore the extension is generated by integral elements.
- (c) Consider the ideal (x, z) . When we quotient we get $\mathbb{C}[y]$ so it is prime. However it is maximal when restricted to $\mathbb{C}[x]$, a contradiction.

9.28

- (a) The image of the map is all the prime ideals of R/P which have an ideal of R'/P' lying over them. These correspond to prime ideals of R' containing P' lying over prime ideal of R containing P . Therefore the conditions are equivalent.
- (b) The image of the map is all the prime ideals in R_P which have an ideal in $R'_{P'}$ lying over them. These correspond to ideal contained in P' lying over ideals contained in P . Then this is equivalent to the lying under property.

Noether Normalization and Hilbert's Nullstellensatz

10.4

We can choose the a_i such that

$$a_1 \gg a_2 \gg \cdots \gg a_{n-1} \gg 1$$

Now consider the lexicographic degree of monomials in $K[x_1, \dots, x_n]$. IE to compare two monomials we first compare their x_1 degree, if they are equal we proceed to the x_2 degree and continue. Now expand the product and consider the highest order term in y_n . This will come solely from the monomial in f with the highest lexicographic degree. Therefore it will be nonzero and monic since there is only one contribution.

In a sense, any choice of the a_i such that their differences are large compared to the degree of f will work.

10.7

We have two polynomials which need to be normalized.

$$xy + z^2 \quad x^2y - xy^3 + z^4 - 1$$

First of all we note that z is already integral over $R[x, y]$. Therefore this algebra is finite over $R[x, y]$. Now sub in $z^2 = -xy$ into the second polynomial to get:

$$x^2y - xy^3 + x^2y^2 - 1 = 0$$

which is the only relation in the subalgebra $R[x, y]$. Therefore take $x' = x + y$ so that $x = x' - y$ and we get a leading order term of $2y^4$ so that y is integral over this algebra. Then $R[x, y, z]$ is integral over $R[x']$ and we are done.

10.8

(a) Right off the bat we may quotient by $\mathfrak{p}_1$ and $\mathfrak{p}'_2$ to reduce to the going down diagram by lemma 9.7a.

$$\begin{array}{ccccc} R' & & \mathfrak{p}' & \subsetneq & \mathfrak{q}' \\ & & \downarrow & & \downarrow \\ R & & \mathfrak{p} & \subsetneq & \mathfrak{q} \end{array}$$

However we do not have the normality condition which is required to use that theorem. Therefore using Noether normalization we may assume that R is integral over $k[x_1, \dots, x_n]$. Then consider the contractions of $\mathfrak{p}$ and $\mathfrak{q}$:

$$\begin{array}{ccccc} R' & & \mathfrak{p}' & \subsetneq & \mathfrak{q}' \\ & & \downarrow & & \downarrow \\ R & & \mathfrak{p} & \subsetneq & \mathfrak{q} \\ & & \downarrow & & \downarrow \\ k[x_1, \dots, x_n] & & \mathfrak{p}^c & \subsetneq & \mathfrak{q}^c \end{array}$$

Firstly, the containment on the bottom row is proper since otherwise $\mathfrak{p}$ and $\mathfrak{q}$ would be incomparable. Also, $k[x_1, \dots, x_n]$ is normal since it is a UFD. Then going down give us some prime ideal $\mathfrak{p}'$ lying over $\mathfrak{p}^c$. By construction it is in between $\mathfrak{p}_1$ (which we quotiented) and $\mathfrak{q}'$.

(b) This is not true in general. For instance consider example 9.26c. If we add a dimension and let $\mathfrak{p}_1, \mathfrak{q}_1 = (z)$ then we obtain the counterexample by reducing to a failure of going down in an integral domain.

Dimension

11.7

Suppose we construct a chain in R/I . The first element in the chain corresponds to a prime ideal containing I . Moreover by maximality of the chain is must correspond to a minimal (isolated) prime ideal containing I . Then the length of this chain is just $\dim R/\mathfrak{p}$.

11.11

Consider the ring

$$\frac{k[x_1, x_2, \dots]}{(x_1, x_2, \dots)^2}$$

which has the ideal $(x_1, x_2, \dots)$ not finitely generated by a vector space argument. However the only prime ideals of this ring of $(x_1, x_2, \dots)$ and 0.

11.16

(a) Suppose we have a chain

$$P_0 \subsetneq P_1 \subsetneq \cdots \subsetneq P_n$$

and let k be the maximal index so that $a \notin P_k$. We would like to replace P_k in the chain $P_{k-1} \subsetneq P_k \subsetneq P_{k+1}$. First we localize at P_{k+1} and quotient by P_{k-1} to obtain R' . By construction the dimension of this ring is 3. Then choose a minimal prime P'_k over (a) . Note that the codimension is at most 1, therefore it cannot be $P_{k+1}R'$ since this has codimension 2. Furthermore it cannot be the (0) ideal (corresponding to P_{k-1}) since it contains a . Therefore it corresponds to a prime ideal containing (a) fitting in the sequence $P_{k-1} \subsetneq P'_k \subsetneq P_{k+1}$. Doing the repeatedly gives a chain of the desired form.

(b) Suppose that $R = \mathbb{R}[x]$. Now construct the chain

$$0 \subset (x)$$

where the right contains x however the left doesn't and we are not allowed to change it.

11.21

Note that the maximal ideals of the quotient will correspond to the maximal ideals containing the quotient. Firstly we can factor the generators of the quotient:

$$(x+y)(x-y) \quad z^2(x-y)$$

Therefore a maximal ideal containing these two will either contain $(x-y)$ or $(z, x+y)$. The former has codimension 2 and the latter (1). Therefore this ring has dimension 1. By Nullstellensatz the maximal ideals are:

$$\mathfrak{m} = (x-a, y-a, z+b) \quad \mathfrak{m} = (x-a, y+a, z) \quad \forall a, b \in \mathbb{C}$$

where the former have height 2 and the latter height 1.

11.22

(a) I recommend to read the proof of Nullstellensatz for Jacobson rings on my website. This problem technically only requires the machinery of the normal Zariski's lemma, however I will use this machinery to coerce the reader into checking out that proof (you owe me). It also makes this proof a lot easier.

Note that $\mathbb{Z}$ is a Jacobson ring therefore by the Zariski's lemma if $\mathbb{Z}[x]/\mathfrak{m}$ is a field then it is finite over $\mathbb{Z}$. Now consider $\mathfrak{p} = \mathbb{Z} \cap \mathfrak{m}$. By the properties of finite (integral) extensions $\mathbb{Z}/\mathfrak{p}$ must be a field so this intersection is a prime number p . Now we can consider a maximal ideal of $\mathbb{F}_p[x]$. This corresponds to an irreducible polynomial as desired.

(b) Suppose that we have a maximal chain of prime ideals in $\mathbb{Z}[x]$.

$$(0) \subseteq \cdots \subseteq \mathfrak{m} = (f, p)$$

and apply exercise 11.16 to assume that all the prime ideals in the middle contain p . However every prime ideal above p is maximal (since $\mathbb{F}_p[x]$ is a PID). Therefore the chain has length at most 2.

11.23

(a) Consider the prime ideals

$$(y) \quad (3)$$

respectively. Note that the only prime ideal of $\mathbb{R}[x]/(x^2)$ is (x) and $\mathbb{R}[y]/(y) \cong \mathbb{R}[x, y]/(x^2 - y, y, x) \cong \mathbb{R}$ as desired.

Note that the only prime ideal of $\mathbb{F}_p[x]/(x^2)$ is (x) and the quotient is isomorphic to $\mathbb{F}_p$ as desired.

(b) Consider the prime ideals

$$(y + 1) \quad (5)$$

respectively. Note that there is no nontrivial prime ideals of $\mathbb{R}[x]/(x^2 + 1)$ since this is a field ($\mathbb{C}$). Therefore the unique prime lying over is (0) but the quotient are not isomorphic.

Note that (3) is not a square in $\mathbb{F}_5$ and therefore $\mathbb{F}_5[x]/(x^2 - 3)$ is a field and the same argument applies. This is not isomorphic to $\mathbb{F}_5$.

(c) Consider the prime ideals

$$(y - 1) \quad (7)$$

respectively. Note the there are two prime ideals of $\mathbb{R}[x]/(x^2 - 1)$ (namely $(x + 1)$ and $(x - 1)$).

Note that in $\mathbb{F}_7$ $x^2 - 3$ factors as $(x - 2)(x - 5)$. Therefore there are two prime ideals lying over.

These names are used in algebraic number theory. Specifically in a Dedekind domain there is a unique factorization of ideals. Therefore if $\mathfrak{p} \in \text{Spec}(R)$ and $R \subset R'$ then the names describe the factorization of $\mathfrak{p}R'$. If it ramifies then the factorization has a repeated factor. If it is inert then it is generated by a larger prime ideal. If it splits then it splits into multiple prime ideals.

11.24

Consider the chain of ideals:

$$I \subseteq (x, y, w + z) \subseteq (x, y, w, z)$$

11.32

(a) Suppose that R has a chain of prime ideals.

$$(0) \subset \mathfrak{p}_1 \subset \mathfrak{p}_2 \subset \cdots \subset \mathfrak{p}_n$$

We claim we can have a transcendence basis of at least this length. Firstly quotient R by $\mathfrak{p}_1$ and by induction we have a transcendence basis $(x_2, \dots, x_n)$ for $R/\mathfrak{p}_1$. We can lift this basis into R and note that they still form a transcendence basis. Now pick some $x_1 \in \mathfrak{p}_1$ and suppose FTSC that there is an algebraic relation in terms of $x_1, \dots, x_n$. When we quotient by $\mathfrak{p}_1$ there can be no surviving monomials, since this would contradict the $x_2, \dots, x_n$ forming a transcendence basis for this ring. Therefore every monomial in this relation must have a factor of p_1 due to primality. However then because this is an integral domain we can remove a factor of p_1 from every monomial. This produces a contradiction since there is nothing stopping us from doing this infinitely, however the monomials must have a finite degree.

(b) Consider the ring $k(x)[y]_{(y)}$. This has transcendence degree (2) since (x, y) form a transcendence basis. On the other hand the krull dimension is (1) since there are no nontrivial prime ideals of $k(y)[y]$ lying inside (y) .

11.33

(a) Note that the product of varieties corresponds to the tensor product of coordinate rings. One can do this problem by constructing transcendence bases a_i and b_i and then taking the basis $a_i \otimes 1, 1 \otimes b_i$. On the other hand justifying that this forms a transcendence basis is a bit tedious. Therefore I propose the alternative solution via category theory nonsense.

Note that $\text{Spec}(-)$ induces an isomorphism of categories between rings and affine schemes (in the

more general sense). Therefore it must be an exact contravariant functor. In particular it turns colimits into limits so that

$$\mathrm{Spec}(A \otimes_R B) \cong \mathrm{Spec}(A) \times_{\mathrm{Spec}(R)} \mathrm{Spec}(B)$$

and the prime ideals of this ring are specified by pairs $(\mathfrak{p}_A, \mathfrak{p}_B)$ such that they agree when contracted to R . In this case $R = k$ is a field and the contraction condition is trivial. Therefore the dimension is $\dim A + \dim B$.

- (b) Construct the variety $X \times Y$. Now note that the intersection of X and Y corresponds to the intersection of $X \times Y$ and the diagonal variety $\Delta = (x, x) \in \mathbb{A}^n \times \mathbb{A}^n$. Working in the coordinate ring $\mathcal{A}(X \times Y)$ note that Δ can be obtained as an intersection of n hypersurfaces.

$$\overline{x_1 - y_1}, \overline{x_2 - y_2}, \dots, \overline{x_n - y_n}$$

where the bar denotes the image in the quotient. Now by Corollary 11.17 (Krull Height theorem) the codimension of Δ is at most n . Since $X \times Y$ is a variety $\mathcal{A}(X \times Y)$ is an integral domain. Therefore by Corollary 11.12 (pure dimension theorem) the dimension of the desired intersection is at least $\dim X + \dim Y - n$.

Valuation Rings

12.12

- (a) Consider the image of an ideal I of R under the valuation map. Suppose that r' has larger valuation than r , and r is in I . Then note that $r'r^{-1}$ has positive valuation and exists in R . Therefore $r'r^{-1}r = r'$ is in the ideal.

If we have a saturated ideal of the value group we can take the preimage into R . These operations are inverse of each other by construction.

- (b) Suppose FTSOC that I and J do not properly contain each other. Then consider $x \in I \setminus J$ and $y \in J \setminus I$. We know that either $\frac{x}{y}$ or $\frac{y}{x}$ is contained in R . Suppose WLOG that $\frac{x}{y} \in R$. Then we have that $x = y \cdot \frac{x}{y} \in J$, a contradiction.

Note that this property of a "total ordering" of prime ideals actually characterizes a valuation ring. Consider $a, b \in R$. We need to show that either $\frac{a}{b} \in R$ or $\frac{b}{a} \in R$. Suppose WLOG that $(b) \subseteq (a)$. Then $b = ac$ and $\frac{a}{b} = \frac{1}{c} \in R$ as desired.

Dedekind Domains

13.18

- (a) Note that fractional ideals are R -submodules of K . Define the product of two fractional ideals as the image of the natural map

$$I \otimes_R J \rightarrow K \quad i \otimes j \mapsto ij$$

and note that localization is an exact functor. Therefore it commutes with the tensor product so that

$$S^{-1}(I \otimes_R J) = S^{-1}I \otimes_R S^{-1}J$$

and by comparing the images of the localized maps we see that

$$S^{-1}(IJ) = (S^{-1}I)(S^{-1}J)$$

- (b) Recall the identity $(I :_K J) = \text{Ann}((I + J)/I)$ in the quotient module K/I . However localization commutes with quotient so we have

$$S^{-1} \frac{I + J}{J} = \frac{S^{-1}(I + J)}{S^{-1}J}$$

and it now suffices to show that localization commutes with taking annihilators (in an integral domain). Note that

$$S^{-1}M = M \otimes_R S^{-1}R$$

and now consider the annihilator of $m \otimes_R 1$. We can see that the orbit under r are elements of the form $rm \otimes_R 1$. Since $S^{-1}R$ is an integral domain this will only be 0 if $rm = 0$ as desired.